

ADJEI BISMARCK NII ADJETEY OTOPAH

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PROFILE

I am an Embedded Systems Engineer with a strong command of the full technology stack. While my foundation lies in designing and programming hardware systems, I have expanded my expertise across backend development, web engineering, data analytics, and graphic design—allowing me to build complete, end-to-end solutions from the hardware level to the user interface. I do not just write firmware; I architect scalable backend systems, analyze the data they generate, and design the platforms that bring them to life. By integrating Edge AI and deploying lightweight machine learning models directly onto resource-constrained microcontrollers, I enable smarter, real-time decision-making at the hardware source. My goal is to bridge the physical and digital worlds through technology that is efficient, scalable, and purposeful.

EDUCATION

BSc. Computer Science *Expected November 2027*
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

W.A.S.S.C.E (General Science) *January 2021 – September 2023*
Our Lady of Grace Senior High School, Mampong, Ghana

B.E.C.E *September 2020*
El-Goshen School Complex

WORK EXPERIENCE

Intern — Electricity Company of Ghana (ECG), Eastern Region (Nsawam)
October 2024 – December 2024

- Assisted in the preparation of statements and reports, ensuring compliance with company policies and regulations
- Supported data cleaning processes and maintained databases
- Conducted customer segmentation analysis

Intern — Smart Hive Labs, Western Region(Prestea)
August 2025 – December 2025

- Integrated Supabase for authentication and storage.
- Designed scalable backend architecture for future SaaS expansion.
- Coordinated with mobile and web teams to ensure shared API usage.
- Handled key business logic including matching, tasks, chat, notifications, and transactions.
- Designed and implemented relational database schema (Users, Skills, Tasks, Messages). Created SQL migrations and ER diagrams using PostgreSQL.
Developed /messages endpoints using WebSockets for realtime chat functionality.
- Designed chat schema and ensured end-to-end communication between clients.
Integrated AI microservice (Flask/FastAPI) into backend for intelligent task matching.
- Built /match endpoint to connect backend services with AI models

Intern — BuzzChat
May 2024 – October 2025

- Assisted in the preparation of marketing collateral and visual assets, ensuring strict adherence to company brand guidelines and design standards.
- Managed and organized the company's digital asset library, maintaining high-quality design files, templates, and multimedia resources.
- Designed targeted visual campaigns and social media graphics tailored to specific customer segments to drive user engagement.

ACHIEVEMENTS / AWARDS

- *Golden Medal of Honor — 1st Place, Claude Hackathon @ KNUST & Anthropic (May 2026)*
- *Certificate of Appreciation, Editor-In-Chief, Computer Science Society, KNUST*

- *Best Student House Chaplain, Our Lady of Grace SHS (August 2024)*
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PROJECTS / RESEARCH EXPERIENCE

Research: Investigating the Effect of COVID-19 in the Educational Sector

- Conducted a comprehensive analysis of the systemic disruptions within the educational sector caused by the pandemic, focusing specifically on the impact of sudden, widespread school closures.
- Evaluated the rapid migration to digital and remote learning platforms, assessing infrastructure readiness, digital literacy gaps, and broadband accessibility in resource-constrained environments.
- Analysed the effectiveness of emergency online learning models against traditional in-person pedagogy, identifying key technical and socio-economic factors that contributed to decreased student engagement and overall learning degradation during the transition.

Research: Boruta-Modified LightGBM for Soil Nitrogen and Phosphorus Prediction

- Conducted advanced predictive modeling using the AfSIS-Ghana spectral dataset to estimate vital soil nutrient levels.
- Applied the Boruta algorithm for rigorous feature selection on high-dimensional spectral data to identify the most significant predictive variables.
- Trained and optimized a LightGBM machine learning model to achieve high-accuracy predictions, demonstrating the application of AI in agricultural data analysis.
- **Objective:** To accurately predict soil Nitrogen and Phosphorus concentrations to support data-driven agricultural decisions.
- **Method:** Developed a machine learning pipeline leveraging the Boruta feature selection technique to filter complex spectral data, paired with a LightGBM algorithm for optimized predictive performance.

Research: FedEdge-Edu (Federated Learning for At-Risk Student Prediction)

Architected a privacy-preserving Federated Learning system to predict at-risk students using localized network interaction data. Deployed FedAvg MLP models on resource-constrained edge nodes, successfully matching the predictive accuracy of centralized models while eliminating the need to transfer sensitive data to the cloud

- Engineered a decentralized Federated Learning (FL) architecture deploying Raspberry Pi 4 edge nodes and MikroTik routers to process educational analytics in low-bandwidth, peri-urban environments.
 - Developed a FedAvg Multilayer Perceptron (MLP) model using PyTorch, utilizing CTGAN for synthetic data generation and SMOTE for local class imbalance mitigation.
 - Achieved $\geq 90\%$ predictive accuracy compared to centralized XGBoost models while ensuring strict data privacy and significantly reducing network bandwidth consumption.
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PROFESSIONAL TRAINING / CERTIFICATION

- **Cloud Computing Certification** — IBM
 - **Data Analytics** — In Progress (Coursera)
 - **Data Analytics** — CognoRise InfoTech
 - **Financial Literacy** — 20th May – 20th June 2024 (Singapore University of Social Sciences)
 - **Certificate of Achievement, Editor-In-Chief** — Computer Science Society, KNUST
 - **Certificate of Completion in Artificial Intelligence** — The Artificial Intelligence(AI) Coordinating Office Awards
 - **Cybersecurity Fundamentals** — IBM
 - **Certificate of Participation** — thedsgnjunkies
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LEADERSHIP EXPERIENCE

- **House Chaplain** — *Our Lady of Grace SHS (September 2022 – September 2023)*
 - **Head of Editorial Committee** — *Computer Science Society, KNUST*
 - **Embedded Project Team Lead** — *Led a technical team in the hardware design and development of a smart gas leakage detector.*
 - **Full-Stack Web Development Team Lead** — *Directed a team of developers in building an AI-integrated web-based project management system.*
 - **Full-Stack Web Development Team Lead** — *Managed a development team to create MindEase, a health-focused web platform designed to support individuals experiencing depression and anxiety attacks.*
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VOLUNTARY / EXTRA-CURRICULAR ACTIVITY

- **First Years Orientation** — Assisted in coordinating student mobility *January 2024*
 - **Rotary Club Donations** — Supported donation logistics *May 2020*
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SKILLS & ABILITIES

- **Programming & Backend:** API Development, RESTful APIs, Authentication & Security, System Architecture, Database Management
- **Data Analysis Tools:** Excel, SQL, R-Language, Pandas, NumPy
- **Data Visualization:** Power BI, Matplotlib
- **Statistical Analysis:** Hypothesis Testing, Regression, A/B Testing
- **Software Practices:** Version Control with Git, Team Collaboration & Code Review
- **Database Design:** Data Modeling & Schema Design
- **Embedded Systems & Hardware:** Arduino Mega & ESP8266 Microcontrollers, Sensor Integration (MQ series gas sensors, OLED displays, relay modules), Circuit Simulation

(Tinkercad, Wokwi), 3D CAD Enclosure Design (Onshape)

REFEREE

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KNUST

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